

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	1.12%	3.61%	3.43%	0.94%
Meat MCal Target	30,541.19	98,698.37	94,015.06	25,857.88
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	480.91	599.53	125.27	407.16
MCals from meat	30541.19	98698.37	94015.06	25857.88
Max Utilized Mcals allowed from Dairy	31869.60	0.00	63584.77	25834.39
MCals actually Utilized from Dairy	31869.60	0.00	63584.77	25834.39
Percent of diet from their meat	1.12%	3.61%	3.43%	0.94%
Percent of Diet from their Dairy	1.16%	0.00%	2.32%	0.94%
Kg Fleece	3,635.23	0.00	0.00	0.00
1 Year+ (age) skins	178.99	23.09	24.23	151.59
Yearly Offspring skins	875.50	706.88	184.85	741.34
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	858.76	109.50	184.21	727.07
Males Step 1	240.45	299.76	62.63	203.58
Females Step 1	240.45	299.76	62.63	203.58
Replacement Mothers per year	172.05	22.20	23.21	145.71
Breeding Males Required	45.08	5.75	9.67	38.17
Additional Mothers	704.64	19.61	87.62	596.75
Males Step 2	197.30	53.68	29.79	167.09
Females 2	197.30	53.68	29.79	167.09
Total Mothers	1,030.81	131.70	207.43	872.79
Total Males	437.75	353.44	92.42	370.67
Total Females	43.15	246.09	32.84	36.49
Total Offspring	875.50	706.88	184.85	741.34
Total Animals	1,556.80	736.98	342.36	1,318.12
Units per Hectare	1.00			
Total Adults per Hectare	2.22	0.34	0.54	2.22
Total Offspring per Hectare	1.09	1.49	0.33	1.09
	0.77%		2.16%	0.65%
Total Adults	1,781	157	305	1,508
Total Offspring	876	707	185	741
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	155.31	0.00	22.58	131.45
Other	14,008.07	0.00	0.00	11,861.49
MCals Dry Mothers	181,398.87	0.00	47,804.71	153,528.67
total Maintenance MCals	195,406.94	0.00	47,804.71	165,390.15

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	521,363	322,727	309,434	441,470
Lactating	994,660	114,727	332,724	842,240
Maintenance	195,407	0	47,805	165,390
Adult Males	52,659	0	20,474	44,584
Offspring	240,326	710,416	676,540	203,499
TOTAL	2,004,415.3	1,147,869.5	1,386,976.3	1,697,182.2

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
Crop	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	875.5	706.9	184.8	741.3
MCals from Straw/Chaff	111889.5	0.0	117077.7	94743.6
MCals from Grain	177289.6	143143.5	74863.0	150121.9
MCals from other feed	289179.1	143143.5	191940.7	244865.5

Diet Calculations		
Plants Mcal	2,342,461.241	
Meat Mcal	273,750	
Dairy	121,288.75864703	
	Perfect	
Plants %	85.57%	
Meat %	10.00%	
Dairy %	4.43%	
check	Perfect	
Percent		
Domestic Meat	9.10%	
Dairy	4.43%	
Wild Meat	0.90%	
Crops	77.01%	
Wild Plants	8.56%	
check	Perfect	
new dairy	121288.76	
potential dairy < limit	TRUE	
new limit	90.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	249,112.500	
Dairy	121,288.759	
Wild Meat	24637.500	
Crops	2,108,215.117	
Wild Plants	234,246.124	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	1031	132	207	873				
Males for Slaughter	438	353	92	371				
Females for Slaughter	43	246	33	36				
Offspring available for Consumption	876	707	185	741				
Gestation Req'd	286379	273715	209694	242465				
Lactation Req'd	546357	97304	225478	462576				
Maintenance Req'd	649985	0	62051	550367				
Adult Males	52659	0	20474	44584				
Offspring Required	287603	774035	809629	243531				
Energy Required	1822983	1145054	1327327	1543523				
Ewe Fleece Kg Yr-1	2319							
Ram Fleece Kg year-1	90							
Juvenile Fleece Kg Yr-1	1226							
Kg Fleece Year-1	3635	0	0	0				